



A Machine Learning-Based Strategy Predicts Selective and Synergistic Drug Combinations for Relapsed Acute Myeloid Leukemia

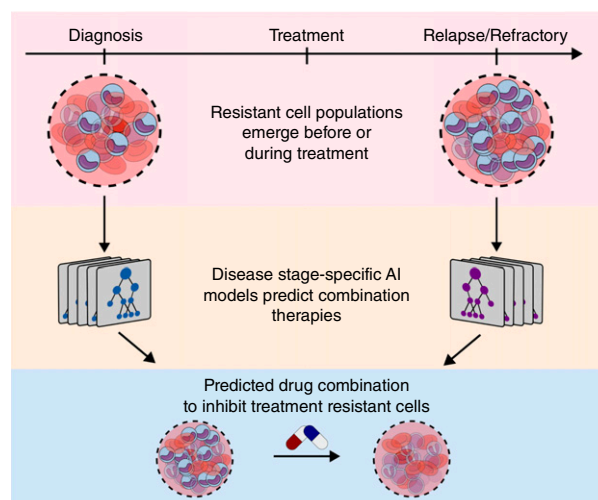
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ABSTRACT

Combination therapies are one potential approach to improve the outcomes of patients with relapsed/refractory (R/R) disease. However, comprehensive testing in scarce primary patient material is hampered by the many drug combination possibilities. Furthermore, inter- and inpatient heterogeneity necessitates personalized treatment optimization approaches that effectively exploit patient-specific vulnerabilities to selectively target both the disease- and resistance-driving cell populations. In this study, we developed a systematic combinatorial design strategy that uses machine learning to prioritize the most promising drug combinations for patients with R/R acute myeloid leukemia (AML). The predictive approach leveraged single-cell transcriptomics and single-agent response profiles measured in primary patient samples to identify targeted combinations that cohibit treatment-resistant cancer cells individually in each sample of patients with AML. Cell type compositions evolved dynamically between the diagnostic and R/R stages uniquely in each patient, hence requiring personalized drug combination strategies to target therapy-resistant cancer cells. Cell population-specific drug combination assays demonstrated how patient-specific and disease stage-tailored combination predictions led to treatments with synergy and strong potency in R/R AML cells, whereas the same combinations elicited nonsynergistic effects in the diagnostic stage and minimal coinhibitory effects on normal cells. In preliminary experiments on clinical trial samples, the approach predicted clinical outcomes of venetoclax-azacitidine combination therapy in patients with AML. Overall, the computational-experimental approach provides a rational means to identify

personalized combinatorial regimens for individual patients with AML with R/R disease that target treatment-resistant leukemic cells, thereby increasing their likelihood of clinical translation.

Significance: A predictive model identifies patient-tailored combinations that cohibit multiple drivers to selectively and synergistically target leukemia cells, which could reduce therapy resistance and enhance treatment outcomes in patients with advanced disease.



This article is part of a special series: Driving Cancer Discoveries with Computational Research, Data Science, and Machine Learning/AI.

Introduction

Molecularly targeted therapies have demonstrated improved treatment efficacy in patients with cancer. However, resistance to

targeted therapies inevitably occurs in most patients, even after an initial response, often leading to relapse and death (1, 2). The biological mechanisms behind therapy resistance remain poorly

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Cancer Res 2025;85:2753–68

doi: 10.1158/0008-5472.CAN-24-3840

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understood for many cancer and treatment types. Moreover, the resistance mechanisms are often patient-specific due to inherent heterogeneity among patients and may evolve during the treatment course (3, 4). In comparison with monotherapies, drug combinations have shown improved therapeutic outcomes, and they enable reduced toxicity and adverse effects as a consequence of reduced effective dosages; importantly, combinatorial therapies can reduce the risk of treatment resistance via coinhibition of multiple cancer- or resistance-driving pathways in the tumor and its microenvironment (5, 6). To identify effective combinatorial therapies for individual patients and to address the combinatorial explosion problem caused by the massive number of potential combinations, computational approaches that make use of transcriptomic data or other molecular profiles have been developed to predict combinatorial responses (7–9).

Compared with bulk-level molecular profiling, single-cell RNA sequencing (scRNA-seq) provides deeper insights into cellular heterogeneity within the tumor microenvironment of an individual cancer, paving the way for a better understanding of the cellular mechanisms contributing to cancer development and therapy resistance (10, 11). Recent studies have demonstrated that distinct cell populations across different stages of leukemia development may lead to distinct response patterns to targeted therapies when investigated both using *ex vivo* response assays and patient clinical outcomes (bioRxiv 2023.12.26.573390; refs. 10–16). For instance, leukemic blast cells of patients with FAB M0/1 acute myeloid leukemia (AML) often demonstrate higher *ex vivo* sensitivity to BCL2 inhibitors, such as venetoclax (12); in contrast, differentiated monocytic cells, more abundantly present in M4/5 AML subtypes, have shown enhanced resistance to BCL2 inhibition (17). Single-cell drug testing studies in AML have also highlighted the importance of using leukemic blast-specific readouts in a flow cytometry-based *ex vivo* drug testing approach to simultaneously evaluate drug sensitivity or resistance in various cell populations of primary AML samples (12, 14, 18, 19).

Earlier scRNA-seq-based computational prediction models have demonstrated their potential for identifying cancer-selective and patient-specific drug combinations that discern cellular heterogeneity and selectively target disease-related cell populations, thereby avoiding broadly toxic combinations that may lead to drug-induced adverse effects (13, 20, 21). However, the current prediction models lack built-in mechanisms for capturing differential contributions of heterogeneous cells to treatment sensitivity and resistance, and they have been developed based on patient samples from a single disease stage (e.g., diagnosis or relapse). This limits their potential to leverage dynamic changes in the cellular composition throughout disease progression and resistance development for predicting combinations targeting resistant cells. Rational strategies are therefore needed for patient-tailored combination designs that use predictive modeling with multistage patient samples. The models should be informed by transcriptomic states inferred from single-cell profiling, toward the discovery of targeted combinations that are safe and effective after relapse occurs.

To fill this gap, we developed a precision medicine approach that leverages both bulk *ex vivo* single-drug sensitivity profiles and scRNA-seq data from longitudinal patient samples to systematically identify combinations of targeted agents for relapsed/refractory (R/R) AML. Based on a cohort of five paired diagnosis–R/R samples, we trained personalized machine learning models individually for each sample to identify targeted drug combinations that (i) show increased synergistic effects in the R/R stage while having

nonsynergistic effects in the diagnostic sample of the same patient and (ii) lead to minimal coinhibition of sample-specific normal cells, thereby increasing their likelihood of better tolerability. The predicted combinations were tailored to inhibit leukemic cells inferred from scRNA-seq data from each patient sample. This is because we revealed extensive cellular heterogeneity within the cohort of patients with AML and dynamic changes in the leukemic and normal cell types between the diagnostic and relapsed stages in response to disease progression and treatment regimens. We show that the computational–experimental approach accurately predicts patient-specific combinations with high synergy and potency for coinhibiting treatment-resistant AML cells.

Materials and Methods

We developed and experimentally validated a systematic strategy to predict personalized drug combinations for patients with R/R AML. Our objective was to identify changes in targetable signaling pathways, identified with scRNA-seq and drug response assays, that took place in the transition from diagnosis to relapse individually in each patient with AML. The experimental–computational approach takes as its input matched scRNA-seq and single-agent screening data from an individual patient sample (diagnosis, relapse, or refractory stages). Computational methods were then used to link scRNA-seq measurements at the single-cell level to *ex vivo* drug sensitivity measurements at the bulk level [CellTiter-Glo (CTG) cell viability assay]. This is because of the scarcity of primary patient cells for single-cell drug response screening, which requires many more cells than the CTG drug screening. As the drug combination responses can vary dramatically between patients and cell lines, even in the same cancer type (20–22), our approach predicts effective and synergistic combinations tailored to each patient's R/R disease state (output of the approach). Experimental validation of the most potent combinations with cell population-specific flow cytometry combination assays was performed in the same patient cells in which the predictions were made.

Paired samples of patients with AML

Bone marrow aspirates were collected from five patients diagnosed with AML who had both diagnosis and relapse/refractory AML samples available from the Finnish Hematology Registry and Clinical Biobank, hosted at Helsinki University Hospital (HUS), with written informed consent from the patients in accordance with the Declaration of Helsinki as part of the individualized systems medicine for AML study (23). Patients' clinical, genetic, and research data are summarized in Supplementary Table S1.

Compound sensitivity testing

Ex vivo single-drug sensitivity screens were performed on freshly isolated cells from bone marrow according to the established protocols and methods implemented at the Institute for Molecular Medicine Finland (23). We used compound collections consisting of up to 544 targeted compounds from the Institute for Molecular Medicine Finland oncology library, which is a comprehensive and evolving compound collection of both approved drugs and investigational compounds. Various versions of the compound library have been used for single-drug testing in patients over the years, depending on when they were diagnosed; therefore, the number of targeted compounds differs between the patient samples (Supplementary Table S1). The *ex vivo* responses in each patient sample to the single drugs were measured at five concentrations using the

CTG cell viability assay (23). The dose–response inhibition data, after normalizing with positive and negative controls on each plate, were fitted by the four-parameter log-logistic function to calculate the drug sensitivity score (DSS). DSS is based on closed-form integration of the area under the dose–response curve, which transforms multiparametric dose–response relationships into a single-response metric for each compound and sample (24). We implemented the dose–response data analysis using the Breeze web application (RRID:SCR_026646; ref. 25). The compounds with well-fitted dose–response data, as evaluated by the Breeze curve fitting options, were used in the predictive modeling.

Compound–target information

We systematically collected compound–target interactions from two public databases: Drug Target Commons version 2.0 (<https://drugtargetcommons.fimm.fi/>; RRID:SCR_026607; ref. 26) and DGIdb version 4.0 (www.dgldb.org; RRID:SCR_006608; ref. 27). Based on our previous study (20), we included protein targets that exhibit a relatively potent dose–response binding affinity (K_d , K_i , $IC_{50} < 1,000$ nmol/L), as sourced from the Drug Target Commons database, and protein targets with a reference score >2 from the DGIdb database. Using these predefined potency thresholds, we identified a total of 1,076 unique protein targets associated with the 544 targeted compounds, including both their nominal on-target and potent off-target activities. The interactions between compounds and protein targets were represented as a binary matrix, in which an element is assigned a value of 1 if the corresponding compound and target interact and 0 if they do not interact.

scRNA-seq data analysis and processing

The patient-derived bone marrow samples were enriched for mononuclear cells using the density gradient-based Ficoll-Paque PREMIUM method (GE HealthCare) before single-cell sequencing using the 10x Genomics (RRID:SCR_023672) Chromium Single Cell 3' RNA-seq platform with Next GEM version 3.1 Dual Index chemistry. Libraries were prepared using the Chromium Next GEM Single Cell 3' Gene Expression version 3.1 Dual Index kit. The library preparation, gel beads in emulsion generation, and cDNA amplification were performed according to the manufacturer's instructions (10x Genomics), aiming for a target cell capture of 4,000 to 6,000 target cell capture per sample. Samples were sequenced on an Illumina NovaSeq 6000 system with read lengths of 28 bp (read 1), 10 bp (i7 index), 10 bp (i5 index), and 90 bp (read 2). Data processing and analysis were performed using 10x Genomics Cell Ranger version 6.0.0 pipelines (RRID:SCR_017344). The “cellranger mkfastq” command, utilizing Illumina's bcl2fastq version 2.2.0 (RRID:SCR_015058), was used to generate FASTQ files from raw base calls. The “cellranger count” pipeline performed alignment against the human genome GRCh38. We applied quality control (QC) to the filtered gene–barcode matrix processed by Cell Ranger from 10x Genomics. QC included removing cells with either a low number of detected genes (below 200) or an unexpectedly high number of cells (3,000–6,000, depending on the sample), using the Seurat R package (version 4.2.0; RRID:SCR_016341; ref. 28). Additionally, cells with more than 10% of mitochondrial unique molecular identifier counts were considered low quality or dying and were excluded. Next, the preprocessed scRNA-seq data were normalized using the LogNormalize method and clustered with the Louvain algorithm implemented in Seurat.

Cell type assignment and blast detection

To classify cell subpopulations in the samples of patients with complex AML, we utilized the scType tool (RRID:SCR_026634; ref. 29) for cell type annotations of single-cell transcriptomes in each patient sample. scType is an in-house developed, freely available web tool that supports fully automated and accurate cell type identification using a comprehensive cell marker database of positive and negative marker genes as background information. To assign each cell type to a cluster, we applied scType to calculate the marker specificity and enrichment scores across cell clusters and cell types, with high scores indicating a high-confidence cell type annotation. scType assigned the “unknown” cell type label to the cell clusters with either a low scType score (less than 20% of the cell number within the cluster) or a negative scType score. To identify undifferentiated AML blast-like cells, we manually added a new cell type, “blast-like cells,” to the scType database, defined by the leukemic blast marker genes (CD34, CD33, CD38, PROM1, ENG, CD99, and KIT; ref. 21) as positive markers and more differentiated markers (CD14 and CD11b) as negative markers. Based on the cell annotations, we calculated the AML blast cell percentage for all the patient samples to characterize the hematopoietic hierarchy of the cell populations.

Predictive modeling of drug combination effects

To systematically identify synergistic drug combinations that target treatment-resistant cancer cells in the R/R samples, we applied an ensemble machine learning model, extreme gradient boosting (XGBoost), that combined *ex vivo* patient-derived single-drug sensitivity profiles, compound–target interaction networks, and scRNA-seq data to predict the responses of drug combinations across cell types. For each AML sample (diagnosis and R/R), the transcriptome-wide expression profile of individual cells was filtered using a subset of genes involved in the compound–target activity profiles. These lower-dimensional single-cell gene expression profiles were then multiplied by a compound–target interaction matrix to compute a matrix of the target-based single-cell enrichment for the compounds, which was utilized as an input to train sample-specific XGBoost models for predicting single-drug response profiles in each sample (the measured DSS profiles). We split the sample-specific model inputs and the measured single-drug DSS profiles into five distinct folds for fivefold cross-validation. Each fold served as a validation set once, whereas the remaining four folds were combined as the training set. The model performance was evaluated computationally by comparing the measured and model-predicted out-of-fold single-drug DSS profiles. By using fivefold cross-validation and grid search, we iteratively trained and evaluated the XGBoost models to identify the most predictive models for single-drug DSS profiles for each sample. To prioritize compounds with high-confidence predictions, we applied conformal prediction (CP; ref. 30) to evaluate the reliability of individual predictions by generating nonconformity scores and subsequently removed single-drug response predictions with a nonconformity score lower than 0.8.

In the next phase, similar to our earlier drug combination prediction and testing platform (31), we constructed a combination–gene interaction matrix, defined as a binary matrix in which the rows correspond to combinations and the columns to drug target genes. For each of the combinations, all the potent drug targets are coded as 1, and the nonactive targets are coded as 0. This drug combination–gene interaction matrix was further multiplied by the scRNA-seq expression matrix and used as input for the best-trained sample-specific single-drug response XGBoost model to systematically predict the unknown sample-specific drug combination

responses (the model output). The predicted efficacy of a combination was determined by the combination DSS predicted by the sample-specific XGBoost model (Supplementary Table S2). The highest single-agent (HSA) synergy model (32) was used to quantify the synergistic effects of drug combinations and the synergy differences between the paired samples, following the independent drug action concept (6). Based on the HSA differences, we identified the drug combinations with increased selective synergy in R/R samples and nonsynergistic effects in diagnostic samples. Specifically, we first selected relapse-specific combinations with predicted HSA synergy scores greater than 5 in the R/R samples while falling between 0 and 5 in the diagnostic samples. Out of the predicted combinations, we further selected the top 10% of combinations with the highest difference in the predicted efficacy between the R/R stage and the diagnostic stage samples.

t-NSE

To identify cell population-specific combinations, we developed a target-based normalized single-cell enrichment (t-NSE) score that quantifies the enrichment in expression of multiple drug targets across cell subpopulations, particularly in cancer cells (here, leukemic blasts and monocytic cells) and normal cells (here, T cells and NK cells). More specifically, for a specific cell type, MinMax scaling was used to normalize the expression of drug target enrichment scores, and the overall enrichment score was calculated by adding the scores of individual cells together and dividing by the square root of the cell number. Given that cell clusters of interest may comprise multiple cell types, we computed the cell cluster-level t-NSE scores by summing the enrichment scores across the cell types and then dividing by the number of cell types.

To compare the target enrichment levels between distinct cell clusters (e.g., cancer cells vs. normal cells), the differential t-NSE scores were calculated by subtracting the t-NSE scores of normal cells from those of cancer cells. Among the predicted relapse-specific synergistic combinations (see the above section), we further identified the selective combinations with the differential t-NSE scores surpassing the median of the differential t-NSE scores across all overlapping combinations between the diagnostic and relapsed samples. These combinations were expected to selectively inhibit cancer cells while minimizing toxic inhibition of normal cells, as indicated by the higher enrichment of their corresponding drug targets in cancer cells when compared with normal cells.

Validation of the selective combination responses

Due to limitations in the availability of patient-derived bone marrow cells, beyond those used in routine diagnostics, we selected four drug combinations that involved seven single agents to experimentally validate the predictions in the paired patient samples of the index patient (AML2 D and AML2 R), for which enough primary cells were available for flow cytometry-based cell population drug combination assays. We tested the four combinations on the bone marrow mononuclear cells of each patient sample in a 4×4 dose-response matrix (including single treatments for both drugs). In contrast to the doses used in the single-drug CTG assays, we focused in the combination experiments on relatively lower concentration ranges, which are often clinically better tolerated. The same three concentrations were chosen for the same compounds in different combinations to make their results more comparable (Supplementary Table S3). The drug combination effects were quantified using the HSA score with the SynergyFinder version 3 web tool (RRID:SCR_026127; ref. 33).

We performed high-throughput flow cytometry assays to assess the coinhibition effects of drug combinations on multiple cell populations in samples of patients with primary AML, similarly to before (20). Briefly, the compounds were dissolved in 100% DMSO and dispensed into conical-bottom 384-well plates (Greiner) either as single agents or combinations in three concentrations using an Echo 650 liquid handler (Beckman Coulter). Cryopreserved bone marrow mononuclear cells were thawed and suspended in HS5 stromal cell line-derived conditioned medium. The cells were seeded on the drug plates at 35,000 viable cells per well with a MultiFlo FX.RAD (BioTek) and incubated for 72 hours at 37°C and 5% CO₂. To profile the cell subpopulation responses, the cells were stained with BV785 Mouse Anti-Human CD14 [RRID:AB_2810577; BioLegend (RRID:SCR_001134)], VB515 Recombinant Anti-Human CD56 [RRID:AB_2658743; Miltenyi (RRID:SCR_008984)], and the following antibodies from BD Biosciences (RRID:SCR_013311): V500 Mouse Anti-Human CD45 (RRID:AB_1937324), BV650 Mouse Anti-Human CD19 (RRID:AB_2744313), PE-Cy7 Mouse Anti-Human CD3 (RRID:AB_396896), PE Mouse Anti-Human CD34 (RRID:AB_393871), BV421 Mouse Anti-Human CD38 (RRID:AB_11151894), and APC Mouse Anti-Human CD117 (RRID:AB_2916458), together with APC/Fire 750 Annexin V (BioLegend) and DRAQ7 (BD Biosciences). The cells were then analyzed with an iQue3 flow cytometer (Sartorius). Remaining live cells after drug treatments were gated using Forecyt (Sartorius). The cell singlets were identified based on forward scatter area versus forward scatter-height (FSC-H) ratio, and live cells were identified by excluding Annexin V- and DRAQ7-positive cells, followed by the identification of leukocytes (CD45⁺). Further characterization was performed for NK cells (CD56⁺CD3⁻), leukemic blasts (CD34⁺ and/or CD117⁺), leukemic stem cells (CD34⁺CD38⁻), monocytes (CD14⁺), and T/B cells (SSC-A_{low} and CD3⁺ or CD19⁺) from the leukocytes.

Turnaround time and costs

The experimental-computational approach takes approximately 2 weeks from sample collection (including scRNA-seq, *ex vivo* CTG drug response assay, the computational tasks, and subsequent validation using the cell population-specific flow cytometry assay), which is a clinically actionable time frame in AML management. The total cost of the protocol is approximately USD 3,200 per sample, but as sequencing costs decrease, the experimental costs are expected to become significantly lower in future.

VenEx trial samples

We analyzed nine samples of patients with AML from the VenEx trial (14). The VenEx trial is a multicenter, two-stage, open-label, phase II trial evaluating the correlation of *ex vivo* sensitivity to venetoclax with clinical outcomes in patients with chemotherapy-ineligible, *de novo*, secondary, or R/R AML treated with venetoclax and azacitidine. To perform the scRNA-seq, freshly frozen mononuclear cells from each patient's bone marrow samples were sequenced using the Chromium Single Cell 3' version 3.1 Dual Index Reagent Kit (10x Genomics). The Cell Ranger version 4.0 analysis pipelines (10x Genomics) were used to preprocess data. The scRNA-seq data were analyzed in the same ways as the real-world AML cohort samples described above. The *ex vivo* drug response for each patient was performed on freshly isolated cells from bone marrow samples. For drug sensitivity testing, 20 mL of bone marrow were shipped from five trial sites to the central laboratory in EDTA tubes at room temperature. Mononuclear cells were isolated using Ficoll

gradient centrifugation and plated in two different drug plates. The drug plates contained 20 compounds, including venetoclax and other BCL-2 family inhibitors across a 10,000-fold concentration range. We excluded one nonargeted drug (idarubicin) from the modeling. After incubation with the drugs for 48 hours, cell viability was assessed in parallel using CTG and flow cytometry drug assays.

Ethics statement

The patient samples and data were collected with written informed consent from the patients, and the studies were conducted in accordance with the Declaration of Helsinki. The real-world study was approved by the Institutional Review Board of the HUS (HUS Ethics Committee Statement 303/13/03/01/2011, latest amendment 7 dated June 15, 2016; latest HUS study permit HUS/395/2018 dated February 13, 2018). The VenEx trial was approved by the Finnish Medicines Agency (KLnro 105/2019, November 2019) and the HUS Ethics Committee (HUS/3235/2019, December 2019, including the translational studies). The hospital agreement with HUS was finalized in January 2020. The trial is registered at the EU Clinical Trials Register (EudraCT 2019-003836-22) and at ClinicalTrials.gov (NCT04267081).

Statistical analyses

The statistical tests applied and the significance values are included in the figure legends or results text. We used nonparametric tests in all the statistical comparisons.

Data availability

The *ex vivo* drug responses of the single compounds tested on the 10 AML samples are available in Supplementary Table S4. The scRNA-seq data of AML5 D and AML5 R samples are available at the European Genome-phenome Archive (EGAS00001004444; ref. 34, and EGAS00001004614; ref. 20). The scRNA-seq data of AML2 R and AML3 R samples are available at the Sequence Read Archive (RRID:SCR_004891; SRX26123587 and SRX26123589; ref. 21). The scRNA-seq data for the remaining samples used in the study (AML1 D, AML1 R, AML2 D, AML3 D, AML4 D, and AML4 R) have been made publicly available at the Sequence Read Archive (SRP535478; https://www.ncbi.nlm.nih.gov/Traces/study/?acc=SRP535478&o=acc_s%3Aa). The clinical outcome and research data from the patients with AML of the VenEx trial are available from the principal investigator of the clinical study upon request (M. Kontro, email: mika.kontro@helsinki.fi). All other raw data generated in this study are available from the corresponding author upon request.

The source codes of the personalized drug combination prediction model are freely available at GitHub (<https://github.com/yingjichen/RR-AML>) and Code Ocean (<https://codeocean.com/capsule/9565667/tree/v1>), with user instructions for installation and usage of the package. Training the model for a single sample using new scRNA-seq and single-drug response profiles takes around 5 hours. Once the XGBoost model has been trained for the patient sample, the open-source prediction method enables one to evaluate 10,000 combinations in 0.1 minutes per patient on a virtual machine with CentOS Linux release 7.9.2009 (core) when using six CPU cores in parallel. The processed Seurat RDS objects of the index AML2 patient's scRNA-seq data (diagnosis and relapse) have been deposited in the GitHub repository as example datasets for running the codes.

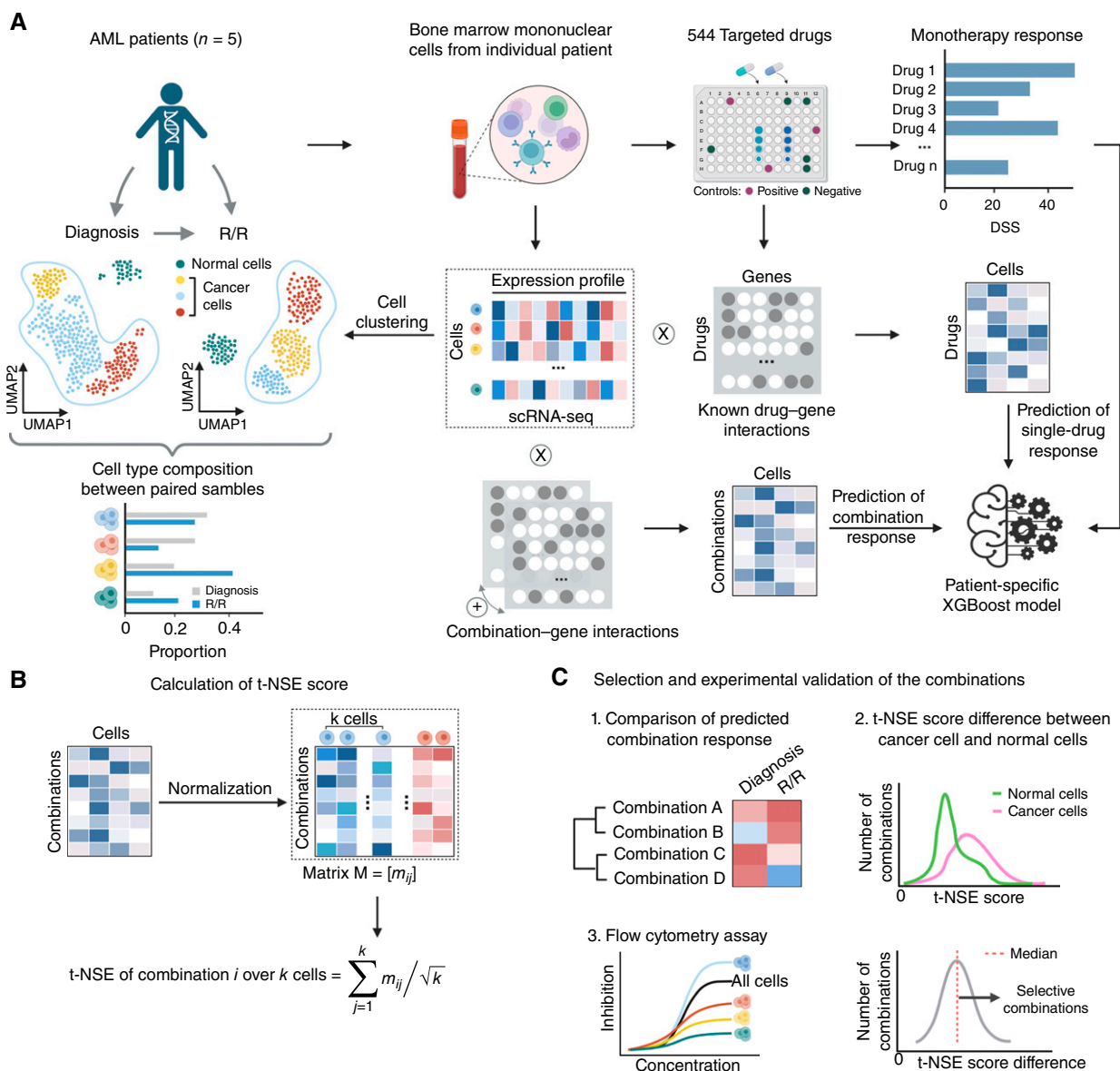
Results

Development of a rational combinatorial design strategy for R/R AML

We tested the approach in a cohort of five paired diagnosis–R/R samples from patients with AML with diverse AML subtypes, genetic driver mutations, and treatment history (Supplementary Table S1). We collected scRNA-seq data at two time points from each patient: at the diagnostic stage prior to the first-line treatment and after the relapse following the first round of treatment and before the next-line treatment. The scRNA-seq datasets across the samples contained expression profiles of 1,504 to 5,651 cells per sample after QC. As a functional input to the approach, we performed drug sensitivity profiling by testing patient samples *ex vivo* against 391 to 466 targeted drugs (CTG readout), depending on the patient sample, with a union of 544 unique agents across the cohort (Fig. 1A). Using this dataset, we developed a predictive approach for cancer-selective combinations in several steps. First, we identified sample-specific cancer and normal cell populations using the scRNA-seq data and positive and negative marker genes in the scType web tool (29). The integration of annotated scRNA-seq profiles and compound-targeted gene interaction networks served as the foundation for training an XGBoost machine learning model separately for each patient (see Materials and Methods) to predict single-agent DSS (24), measured with a bulk-level drug response viability assay that is routinely performed in our individualized AML study (23).

To capture differences in single-cell expression patterns between various cell populations, we implemented a t-NSE score. The t-NSE score quantifies the enrichment in the expression of drug targets in each cell subpopulation, enabling the XGBoost model to connect the drug responses to target gene expression in the heterogeneous AML samples via the integration of compound–target interaction networks and scRNA-seq profiles (Fig. 1B). We used the trained XGBoost models to predict personalized and sample-specific combination responses using the combinatorial t-NSE score, obtained by combining the potent targets of both drugs in the combination. Importantly, t-NSE scores make the models biologically more interpretable by elucidating polypharmacological mechanisms for selective drug combination effects and identifying differential expression of target genes in cancer and normal cell types that may explain individual differences in the combination responses. We also applied CP to exclude low-confidence predictions of the single-drug responses when predicting the patient-specific combination responses. Together, XGBoost, t-NSE, and CP provide a quantitative framework for identifying confident, explainable, and potent drug combination candidates individually in each patient sample.

In the final step (Fig. 1C), we used differences in the combination response predictions between diagnosis and R/R samples and in the t-NSE scores between normal lymphoid and cancer cells to identify unique drug combinations for each patient that show (i) increased combination synergy in the R/R sample and nonsynergistic effects in the diagnostic sample (i.e., patient-specific combinatorial vulnerabilities after treatment resistance) and (ii) selective coinhibition of cancer cells and minimal toxic inhibition of normal lymphoid cells in the relapsed sample (i.e., less toxic combinations that may be better tolerated in the clinics). The top combinations were experimentally tested *ex vivo* using flow cytometry cell population-level drug combination assays to measure relative coinhibition of cancer and normal cells as induced by the predicted combinations in the same cells in which the predictions were made. The coinhibition effects on normal cells provide a preclinical proxy for potential treatment toxicity, which can be

**Figure 1.**

An experimental-computational approach for predicting and validating targeted drug combinations. **A**, Identification of cancer cell-selective and synergistic drug combinations for each patient sample by combining drug-target interaction networks, single-cell gene expression profiles, and *ex vivo* single-drug sensitivity data. We developed and tested the pipeline using *ex vivo* responses to 544 targeted compounds measured for the bulk mononuclear cell fraction using the CTG cell viability assay in a cohort of five paired diagnosis-R/R AML patient samples. The computational search prioritized the most synergistic combinations by using both potent on- and off-targets of the drugs to narrow down the combinatorial search space of more than 75,000 pairwise combinations, far beyond what could be tested experimentally in scarce patient cells. A matrix multiplication algorithm was applied to integrate scRNA-seq data of each sample with drug-gene interactions to generate the target-based enrichment profiles of compounds as input for the sample-specific XGBoost predictive models. We considered the union of targets of the drugs in a combination as the target profile of the drug combination and used the combination-gene target expression score as input for the XGBoost models to predict the combination responses across the paired samples with distinct cell type compositions. UMAP, Uniform Manifold Approximation and Projection. **B**, The t-NSE score was developed to quantitatively compare the coinhibition effects of different combinatorial strategies among various cell types using the scRNA-seq data of each patient sample (see Materials and Methods). We hypothesize that the combinations with higher t-NSE score differences, indicating higher target gene enrichment in cancer cells than in normal cells, will selectively coinhibit leukemic cells while minimizing inhibition of normal cells. **C**, Selection of the patient-specific and cancer cell-selective combinations for the R/R samples. The differential synergy analysis prioritized drug combinations that showed higher predicted synergy in the R/R sample when compared with the diagnostic sample of the same patient (step 1). The target expression selectivity analysis identified combinations that showed higher t-NSE score differences between cancer and normal lymphoid cells, relative to the median scores across all the overlapping combinations between the paired samples (step 2). The cell population-specific *ex vivo* sensitivity and synergy of the selected combinations were experimentally evaluated using flow cytometry drug combination assays (step 3). Created in BioRender. Chen, Y. (2025) <https://BioRender.com/1970449>.

used to prioritize potentially better tolerated combinatorial regimens. Taken together, the computational–experimental approach identifies, for each R/R patient, safe and effective combinations preclinically tested in scarce patient-derived cells.

Prediction of monotherapy responses across diagnosis and relapse/refractory stages

We first trained and tested the sample-specific XGBoost models using the scRNA-seq data and *ex vivo* single-drug response profiles of up to 466 anticancer compounds with various mechanisms of action (Fig. 2A); the single-drug responses were measured in each patient sample using the CTG bulk viability assay (see Materials and Methods). We observed relatively high and statistically significant correlations between the model-predicted and experimentally measured *ex vivo* drug responses in each patient sample (Pearson correlations 0.507–0.601, $P < 2.2 \times 10^{-16}$; Fig. 2B). We did not observe any significant differences in the prediction accuracies between the diagnosis and R/R samples. As expected, the accuracy improved further when considering only the high-confidence predictions informed by the CP (80% confidence; correlations, 0.712–0.843; $P < 2.2 \times 10^{-16}$). Such a high prediction accuracy was rather surprising given the wide heterogeneity observed in the *ex vivo* response profiles, both between the patients and between the diagnosis and R/R stages (Supplementary Fig. S1A–S1F).

There was no apparent enrichment of unreliable predictions in any of the drug or target classes; instead, the CP improved the accuracy of predictions across the drug classes, as exemplified for AML2 samples (Fig. 2C). The monotherapy response predictions for the four other patients showed similarly high accuracies across drug and target classes after the CP (Supplementary Fig. S2A–S2D). The CP removed relatively higher proportions of active drugs (DSS >5) at an 80% confidence cutoff in 9 out of 10 samples ($P < 0.05$, Fisher's exact test), suggesting that drugs with zero or low DSS are more confidently predicted in most samples (i.e., drugs that do not target driver oncogenes of the individual cancer; Fig. 2D). However, the XGBoost models used the whole spectrum of drug responses, both inactive and active drugs in each patient sample, provided their single-agent responses were confidently predicted (Supplementary Fig. S3).

In the following results, we use AML2 as the index patient to showcase the patient-specific prediction and validation results. As an example, the responses to the BCL2 inhibitor venetoclax were reliably predicted in the index patient AML2 samples (Fig. 2C), and across the patient cohort, the sample-specific XGBoost models enabled accurate prediction and ranking of the patients in terms of their *ex vivo* venetoclax response differences between the diagnosis and R/R samples (Supplementary Fig. S4A). Interestingly, both the model predictions and experimental measurements showed that venetoclax monotherapy was not highly effective *ex vivo* in R/R cells of these patients (Supplementary Fig. S4B). The same applied to many other confidently predicted drugs in patient AML2; for instance, those that are labeled in Fig. 2C and that showed relatively low single-drug responses, yet they were later predicted to lead to combination synergies. This suggests that combination therapies are required in the R/R stage, potentially using inhibitors other than those targeting BCL2.

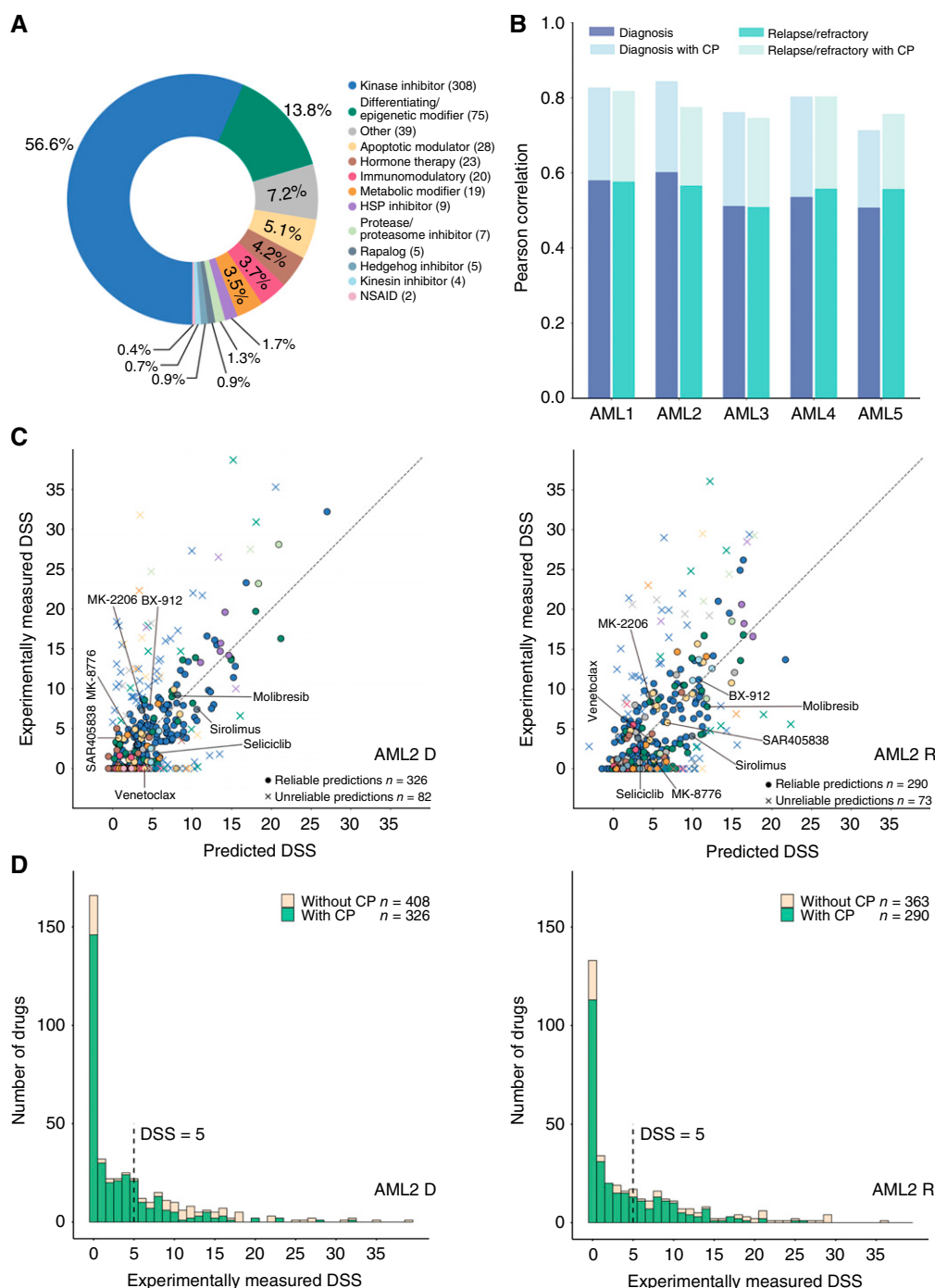
Cellular composition differences between diagnosis and relapse/refractory stages

We next investigated the dynamic changes in normal and leukemic cell populations during disease progression and treatment

regimens and how these contribute to the model-predicted and *ex vivo*–measured drug responses in each patient. Using the single-cell transcriptomic profiles of each AML sample, we mapped the cellular composition of samples at diagnosis and R/R stages. The index patient AML2 is classified as the FAB M5 subtype; the leukemic cells of this patient had become resistant *in vivo*, yet the patient responded to the cytarabine–idarubicin induction therapy (Supplementary Fig. S5). As expected for a patient with the FAB M5 subtype, the scRNA-seq data showed that the diagnosis and relapse stage samples had an increased proportion of monocytic cells compared with those of the non-M5 samples ($P < 10^{-10}$, one-tailed Fisher exact test; Fig. 3A). Interestingly, we further observed a higher proportion of monocytic cells in the AML2 relapsed sample compared with the diagnosis stage sample ($P = 0.0007$, one-tailed Fisher exact test). Aberrant monocytic cell populations in bone marrow have been reported to promote the expansion of leukemia stem cells in cytarabine-treated patients with AML (35, 36). Such leukemia stem cell expansion may have similarly contributed to the relapse of the AML2 patient after an initial response to the induction therapy.

We also observed a higher proportion of dendritic cells (DC; $P < 10^{-10}$, one-tailed Fisher exact test) and macrophages ($P < 10^{-10}$) in the diagnosis sample, as compared with the relapse stage sample of patient AML2 (Fig. 3A). DCs are versatile antigen-presenting cells that interact with normal naive T cells, inducing their activation, differentiation, and polarized effector response, hence serving as a crucial link between innate and adaptive immunity and triggering specific cellular responses against cancer. It has been reported that the immature, defective, or reduced number of DCs results in the unresponsiveness of T cells and immunosuppression in the tumor microenvironment, indicating disease progression and leukemia relapse (37). The decreased DC levels in the relapse stage of AML2 may, therefore, be linked to disease progression and higher immunosuppression after the relapse, as compared with the diagnostic stage. Similarly, the higher enrichment of M1-like macrophage markers observed in the diagnosis sample of AML2, when compared with the M2 macrophage markers (Supplementary Fig. S6), may prognosticate a prolonged response to the induction therapy, as M1 macrophages have been shown to induce anti-tumorigenic activity in AML (38, 39).

In each of the patient samples, we observed an increased proportion of leukemic blast cells in the R/R sample compared with the diagnosis stage ($P = 0.031$, one-tailed Wilcoxon signed-rank test), whereas similar changes were not observed in the normal lymphocytic cells (Fig. 3B). This indicates that treatment resistance may lead to higher proportions of AML cell types, which could be selectively inhibited by synergistic drug combinations, predicted separately for each patient and leukemic cell population based on the single-drug response and single-cell transcriptomic profiles using the XGBoost model. To identify undifferentiated AML blast-like cells, we used the existing and newly added marker genes in the scType tool (29). Specifically, a new cell type, “blast-like cells,” was added to scType, defined by the leukemic blast marker genes (CD34, CD33, CD38, PROM1, ENG, CD99, and KIT; ref. 21) as positive markers and more differentiated markers (CD14 and CD11b) as negative markers. Neither the AML2 diagnostic nor relapse samples expressed CD34 (Supplementary Fig. S7), and the cell clusters with increased expression levels of other blast markers (e.g., CD38 and CD33) and a lack of more differentiated marker expression were identified as blast-like cells using scType, which were further annotated as AML blasts (black dashed lines, Fig. 3C and D).

**Figure 2.**

Prediction of monotherapy responses in patient samples using the XGBoost model. **A**, Classification of 544 targeted compounds by their mode of action, with the number of compounds in each class indicated. **B**, Prediction accuracy in terms of the Pearson correlation coefficient between the experimentally measured and model-predicted DSS of monotherapy responses in the paired AML samples without/with CP at 80% confidence. The monotherapy responses were measured using the CTG whole-well cell viability assays. **C**, Correlation between experimentally measured and model-predicted out-of-fold DSS levels in the diagnostic sample (left) and relapsed sample (right) of the index patient AML2. Compounds with unreliable predictions were removed using CP (80% confidence), and the remaining compounds were used in the patient-specific prediction of combination responses. The colors correspond to drug and target classes in **A**. Example compounds are labeled, including the BCL2 inhibitor venetoclax, the mTORC1 inhibitor sirolimus, the BET inhibitor molibresib, and the Akt inhibitor MK-2206, which showed relatively low single-drug responses but were later shown to lead to combination synergies. **D**, Distribution of experimentally measured DSS values for all compounds and those that passed the 80% confidence of CP in the diagnostic sample (left) and the relapsed sample (right) of the index patient AML2. The responses below the dotted vertical line (DSS <5) are generally considered nonactives yet are still used in the XGBoost model for the patient-specific prediction of combinatorial responses, provided they were predicted reliably.

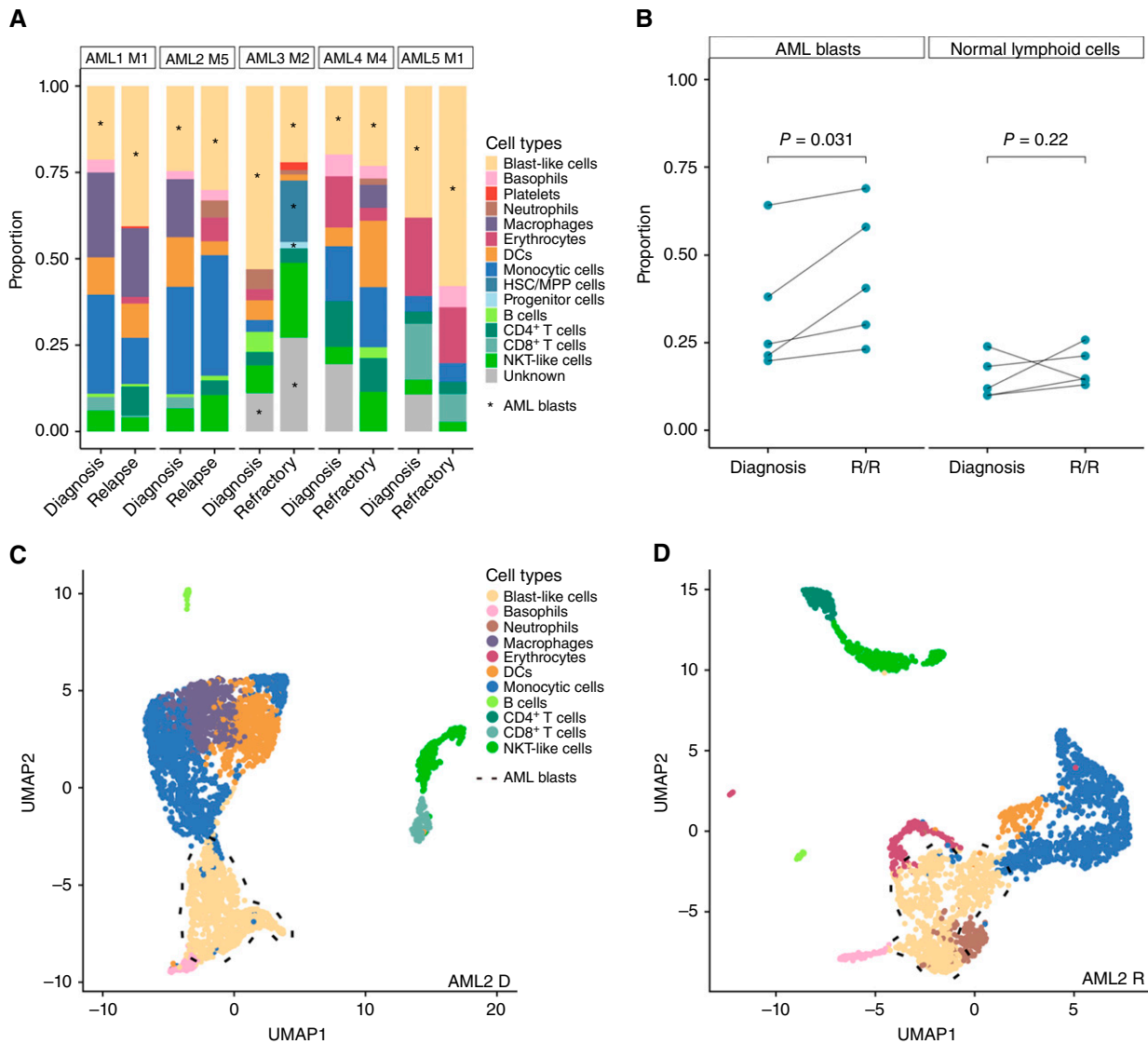


Figure 3.

Inter- and intrasample differences in longitudinal single-cell transcriptomic profiles. **A**, Cell type compositions across the five paired AML samples, identified using the scType tool with newly added marker genes. HSC/MPP, hematopoietic stem cell/multipotent progenitor; NKT, natural killer T. *, AML blasts. **B**, The proportions of scRNA-based leukemic and normal lymphoid cells between the diagnostic and R/R stage samples. Statistical differences were assessed with a one-tailed Wilcoxon signed-rank test. **C** and **D**, Cell type annotations of the index patient AML2 samples: diagnostic (**C**) and relapsed (**D**). In each patient sample, we considered T and NK cells as normal lymphoid cell types and AML blasts and monocytic cells as cancer cells. Leukemic blast-like cells were identified with the scType tool using CD34, CD33, CD38, PROM1, ENG, CD99, and KIT as positive markers and CD14 and CD11b as negative markers. The black dashed lines indicate the cell populations identified as AML blasts in the paired samples. Cells that were annotated neither as cancer cells nor normal lymphoid cells were excluded from the further analyses and combination predictions. UMAP, Uniform Manifold Approximation and Projection.

As noncancerous normal lymphocytic cell types in AML, we used CD4⁺/CD8⁺ T cells and NK cells based on the previous report (40), both identified using the available cell type markers in the scType tool. Additionally, cell clusters with increased CD34⁺ expression levels (for the CD34⁺ samples) and blast-like cells, as identified with scType, were classified as undifferentiated AML blasts (black dashed lines, Supplementary Figs. S7 and S8). The cell type classifications in the paired samples revealed excess heterogeneity across the patients and between the diagnosis and R/R stages (Supplementary Fig. S8), supporting the need for

personalized combination predictions targeting sample-specific cancer cells.

Prediction of selective and synergistic combinations for R/R AML

To combinatorially target the treatment-resistant leukemic cell populations of an individual patient, we predicted synergistic drug combinations that coinhibit the cancer cells unique to each patient sample in the R/R stage using the disease stage-tailored XGBoost models (see Materials and Methods). As drug combinations often

modulate their cellular effects through multiple protein targets, leading to distinct target expression signatures across cell types, we used drug–target networks and scRNA-seq data to guide the combinatorial search process among the thousands of possible combinations. To exploit the target expression variability across cell types and disease stages, we developed the t-NSE scoring, which supported the identification of combinations selectively targeting mainly cancer cells while minimizing the coinhibitory effects on normal cell populations.

In the index patient AML2, the diagnosis and relapsed samples had clearly distinct t-NSE signatures for the selected set of drug combinations (Fig. 4A). More specifically, cancer cells (leukemic blasts and monocytic cells) showed relatively higher t-NSE scores when compared with the normal cell types, especially in the relapse sample, indicating a higher expression enrichment of the combinatorial targets in the cancer cells. This contributes to the selective coinhibition effects of the combinations on the resistant cancer cells. When studying the expression of all single cells, the two drugs in the combinations showed orthogonal (noncorrelated) patterns of target expression profiles, both in normal lymphoid and cancerous cells (Fig. 4B), which likely contributes to the synergistic coinhibition effects of the combinations targeting complementary cancer pathways.

As the combination predictions were based on the XGBoost models initially trained for single-drug response prediction, we also retrieved the built-in feature importance from the trained XGBoost models to evaluate the contribution of each cell and cell type to the monotherapy response predictions across patients and samples. In the index patient AML2, we observed that the cells most critical for predicting drug response were primarily from the cancer cell types, particularly AML blasts and monocytic cells (Fig. 4C and D). This is in line with the higher proportion of monocytic lineage in the AML2 samples and demonstrates the diverse expression patterns among the cancer cells important for cancer-selective drug response prediction. However, the predictive importance of cells and cell types varied across the patients and between the diagnosis and R/R stage (Supplementary Fig. S9A–S9H).

Selective drug combinations show relapsed stage-specific coinhibition effects

For each patient, we selected the overlapping drug combinations between the diagnosis and the R/R stages, as predicted by the XGBoost model among more than 22,000 possible combinations, and ranked them in descending order based on the predicted synergy in the R/R sample to focus on potent combinations in advanced disease (Fig. 5A). Differential t-NSE scores between cancer cells (AML blasts and monocytic cells) and normal lymphoid cells (T cells and NK cells) enabled us to further prioritize those combinations with higher t-NSE scores among the drug combinations that were predicted to synergistically coinhibit resistant cells in each of the AML samples. This led to a total of 24 resistant cell-targeting combinations in the index patient AML2 (Supplementary Table S2), which included a wide variety of target mechanisms, among which we selected combinations with relevance and novelty in AML.

We experimentally validated four targeted combinations among the top-predicted selective combinations in the index patient AML2 using the multidose flow cytometry assay, which quantifies their coinhibition effects on cancer and normal cells separately at various doses (Supplementary Table S3). Experimental validations were carried out only in this patient because there were enough bone marrow primary cells available for *ex vivo* response testing from the patient AML2 diagnostic and

relapsed samples. The replicate flow experiments showed high technical reproducibility for monotherapy and drug combination responses, both in normal lymphoid cells and cancer cells (Supplementary Fig. S11A and S11B). We quantified the synergistic effects on the cancer and normal cells of each of the combinations based on the HSA synergy score, following the independent drug action concept (6).

Notably, the model-predicted HSA synergy correlated with selective coinhibition of cancer cells across diagnosis and relapse when focusing on the combination effects measured at the doses leading to maximal synergy in the relapsed sample of patient AML2 (Fig. 5B). This suggests that the model can identify resistant cancer cell-selective synergies. The average HSA score over the tested 4×4 dose–response matrix demonstrated that the two sirolimus combinations (either with MK-2206 or SAR405838) induced a clear relapse specificity in their selective synergy against cancer cells over multiple and relatively low doses (Fig. 5C). In the two other predicted combinations (molibresib + MK-8776 and seliciclib + BX-912), the differential HSA synergy between the diagnosis and relapse stages was not so large even though the relapse samples showed somewhat wider synergistic areas (Fig. 5C).

When focusing on the drug concentrations that led to the maximal synergistic effect in the relapse sample of the index patient, we observed markedly enhanced cancer cell-selective synergistic effects (i.e., the difference in the coinhibition effects between leukemic and normal lymphoid cells) in the relapsed sample (Fig. 5D). Notably, the selective inhibition of cancer cells increased in each of the four combinations when compared with the corresponding monotherapies (dose-specific HSA of 24.7, 5.0, 12.2, and 20.6, respectively), which indicates a strong synergistic effect, whereas there was no similar increase in the combinatorial inhibition of normal cells, neither in the diagnosis nor relapse, indicating nonsynergistic toxic effects. This suggests that the model can identify preclinically safe combinations that show selective synergy in the resistant cells of the patient in the R/R stage.

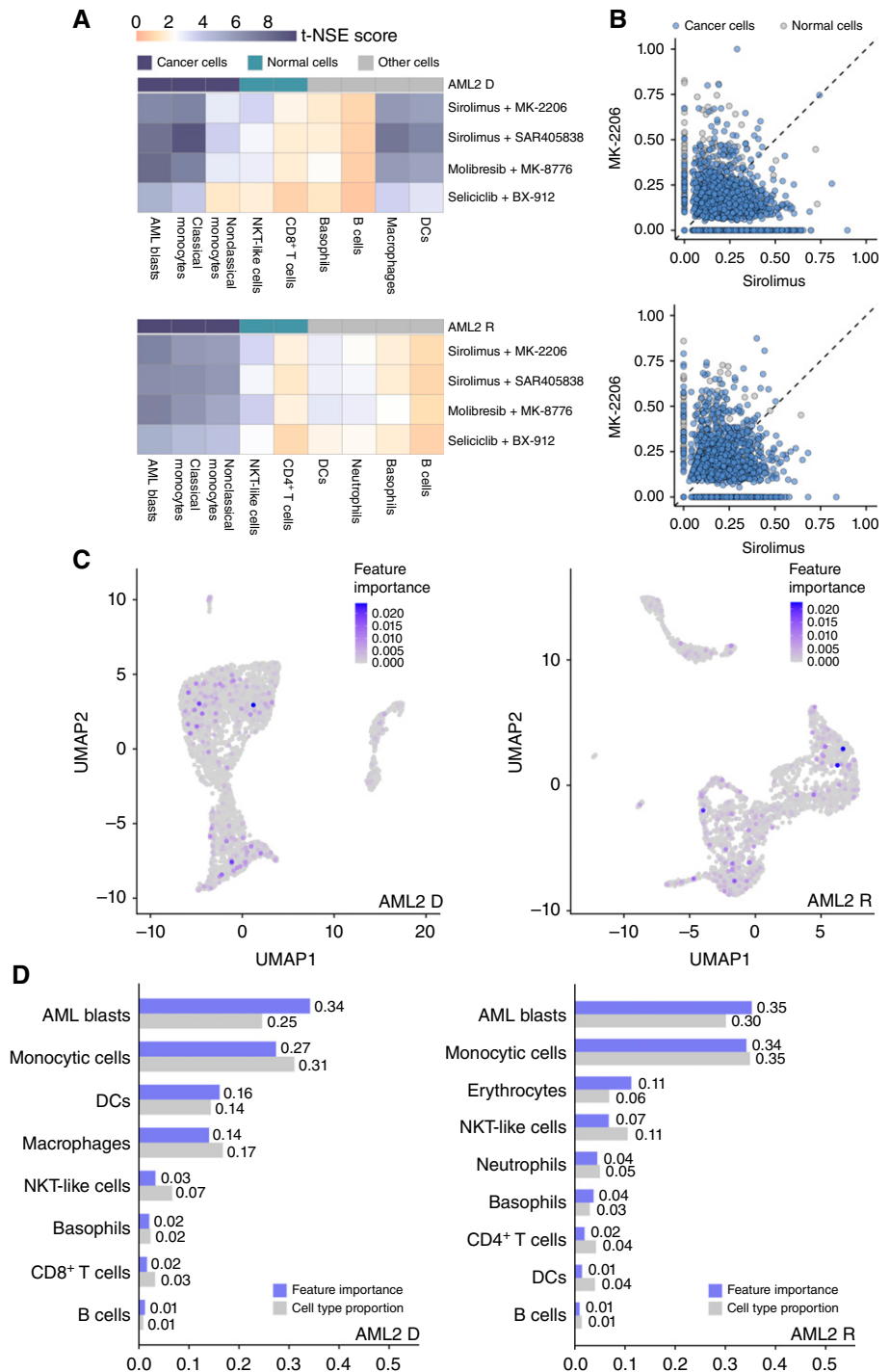
Prediction of patient clinical responses to venetoclax–azacitidine combination therapy

To test the predictive accuracy of the approach in terms of clinical treatment responses, we used the scRNA-seq data and patient outcomes from the first stage of the prospective phase II VenEx clinical trial that evaluates the utility and predictive value of venetoclax *ex vivo* sensitivity testing in patients receiving venetoclax–azacitidine combination therapy (14). Participants with *de novo* AML ineligible for intensive chemotherapy, R/R AML, or secondary AML were included in the study. The primary endpoint was the treatment response in participants showing *ex vivo* sensitivity to venetoclax when using a flow cytometry drug assay. We predicted the *ex vivo* and *in vivo* response to the venetoclax–azacitidine combination therapy in nine samples of patients with AML treated in the VenEx clinical trial with scRNA-seq data currently available. Out of the nine samples, five were from AML cases classified as clinical nonresponders to venetoclax–azacitidine, and four were from clinical responders, based on the objective response rate and the European LeukemiaNet for AML guidelines (41).

For each VenEx AML patient sample, we trained the sample-specific XGBoost model using bulk-level *ex vivo* responses to 19 targeted compounds available from the study, together with scRNA-seq profiles of the samples, to predict the single-agent treatment responses, using the same computational pipeline illustrated in Fig. 1A. The integration of scRNA-seq profiles and potent

Figure 4.

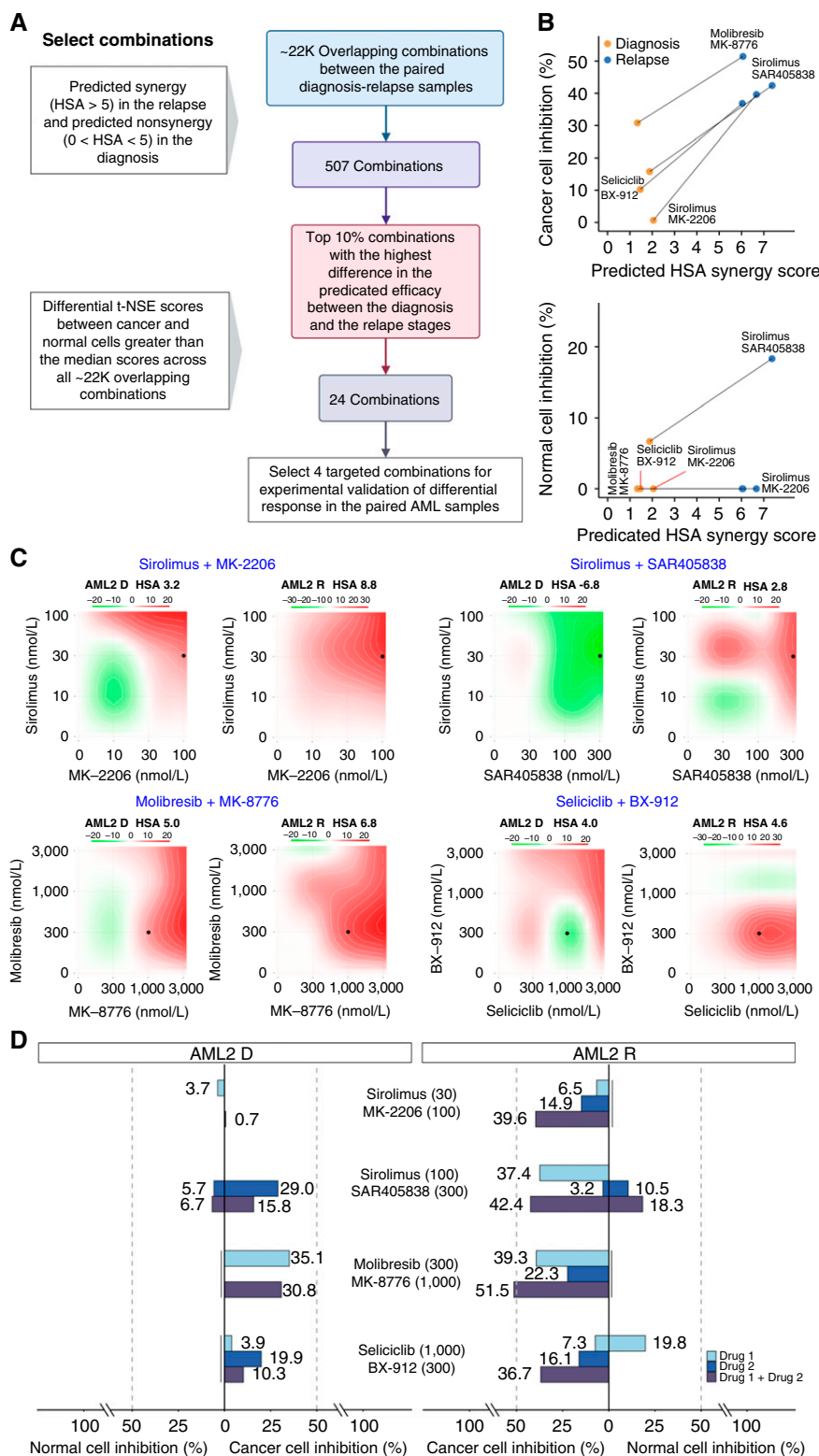
Single-cell transcriptomics contribute to cancer-selective drug response predictions. **A**, t-NSE scores calculated based on scRNA-seq profiles and drug-target networks across cell types in the diagnosis and relapse samples of the index patient AML2 for the selected drug combinations. Note: The other cells were not used in the differential synergy analyses. **B**, The min-max normalized target enrichment scores across cancer and normal cells between the two drugs in an example combination in the AML2 diagnostic (top) and relapsed (bottom) samples. Each point corresponds to a single cell. Supplementary Fig. S10A and S10B show similar scatterplots for the other three combinations. **C**, Uniform Manifold Approximation and Projection (UMAP) visualization of the built-in feature importance for each cell in the XGBoost models that predict single-drug response in the paired samples of the index patient AML2: diagnostic (left) and relapsed (right). The cells with higher feature importance are marked in darker blue, and these features were more important for generating the model predictions for single-drug responses. **D**, The blue bars show the accumulated feature importance that indicates the overall predictive contribution of each cell type to the XGBoost model predictions, separately for diagnosis (left) and relapse (right), calculated by summing up the feature importance within each cell type. The gray bars represent the cell type proportions annotated from the scRNA-seq data using the scType web tool with the newly added marker genes. NKT, natural killer T.



targets of the venetoclax–azacitidine combination (BCL2, BCL2L1, MCL1, IDH2, DNMT1) was used as the input for the trained XGBoost models to predict the combination response to venetoclax–azacitidine in terms of the HSA synergy. Even though the compound panel included only 19 targeted drugs relevant for AML treatment (such as venetoclax, navitoclax, azacitidine, sorafenib, dasatinib, midostaurin, gilteritinib, dexamethasone, and

idasanutlin), the patient-specific models were predictive of the *ex vivo* responses to the narrower target space (Fig. 6A and B), with an accuracy comparable with the prediction of single-agent responses using the larger compound library (Fig. 2B).

When comparing the HSA synergy predicted by the models with the clinical responses of the patients to venetoclax–azacitidine treatment, we observed a higher predicted synergy

**Figure 5.**

Experimental validation of the predicted combinations in paired AML2 samples. AML2 was used as an index patient because there was an insufficient amount of primary bone marrow cells available from the other four patients for the flow cytometry-based experimental validations. **A**, Selection of the four relapse-specific and cancer-selective combinations for the flow cytometry experimental validation in the index patient AML2 (Supplementary Tables S2 and S3). The HSA synergy model was used to score synergistic combinations with a cutoff of HSA > 5. The median t-NSE score level calculated over all the overlapping >22,000 combinations between the paired samples was used as the threshold for selective combination selection. **B**, Relationship between the predicted HSA synergy score and the measured coinhibition of cancer cells (top) and normal lymphoid cells (bottom) of the combination effects measured at the doses leading to maximal synergy in the relapsed sample of patient AML2 (shown as black points in **C**). Note that the y-axis has different ranges in the two plots. **C**, Synergy distributions show the coinhibition of cancer cells in the paired samples. HSA indicates the average highest-agent synergy score over the 4 × 4 dose-response matrix. The multidose-response matrix data from the flow cytometry combination assay are shown in Supplementary Fig. S12. **D**, Measured inhibition effects on normal and cancer cell populations. AML blasts and monocytic cells were used as cancer cells, whereas T cells and NK cells were used as normal cells in the flow cytometry assay. The numeric values in parentheses indicate the drug concentrations used in the treatment experiments (nmol/L). The dashed vertical lines represent the 50% inhibition of normal lymphoid or cancer cells. Each sample was screened in two replicates, and the bars show the average inhibition levels. Supplementary Fig. S11 shows the correlation between the technical replicates of the flow cytometry experiments. Supplementary Fig. S13A–S13H show the expression differences of protein targets of the selective combinations between the normal lymphoid and cancer cells. **A**, Created in BioRender. Chen, Y. (2025) <https://BioRender.com/74oz4jd>.

for the responder group compared with the nonresponder group ($P = 0.032$, two-sided Wilcoxon rank-sum test; **Fig. 6C**). In comparison, the bulk drug responses to neither of the drugs were

predictive of the patient outcomes (**Fig. 6D**). These results demonstrate that our integrated approach predicts clinical outcomes of patients with AML to the venetoclax–azacitidine

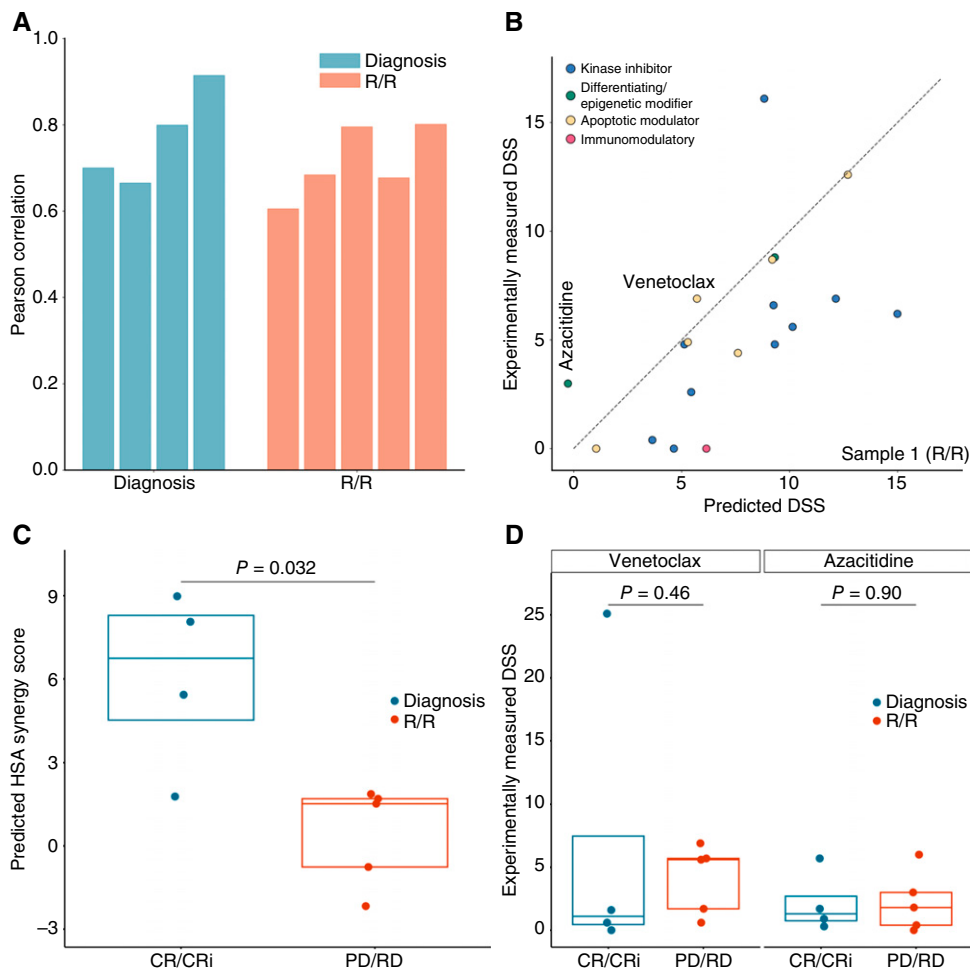


Figure 6.

Prediction of responses of patients with AML to venetoclax-azacitidine combination therapy. **A**, Prediction accuracy in terms of the Pearson correlation coefficient between the experimentally measured and model-predicted out-of-fold DSS of monotherapy responses in the nine AML samples from the VenEx trial with scRNA-seq data available without CP. The monotherapy responses to 19 targeted compounds were measured using the CTG whole-well cell viability assays. **B**, Scatterplot between the experimentally measured and model-predicted out-of-fold DSS profiles of monotherapy responses across the 19 targeted compounds tested in the VenEx trial in an example patient sample (the rest of the patient samples are shown in Supplementary Fig. S14). **C**, Predicted HSA synergy scores in the responder patients ($n = 4$, who achieved CR or CRI to the venetoclax-azacitidine combination treatment) vs. nonresponders [$n = 5$, experiencing refractory disease (RD) or progressive disease (PD) after the treatment]. We defined the patients with progressive disease and refractory disease as nonresponders and patients with complete remission (CR) and complete remission with incomplete hematologic recovery (CRI) as responders. **D**, The single-drug sensitivity scores from the CTG assay, neither for venetoclax nor azacitidine, were predictive of the patient outcomes. Statistical significance was assessed with the two-sided Wilcoxon rank-sum test.

combination, which is currently being widely used for the treatment of *de novo* AML in patients ineligible for intensive chemotherapy (42). This result supports the predictive value of *ex vivo* drug responses for combination therapy optimization, in line with several recent studies that have shown how *ex vivo* single-agent responses correlate with patients' clinical outcomes, both in hematological malignancies and solid tumors, especially when using single-cell drug assays such as those based on flow cytometry or imaging technologies (19, 23, 43–47).

Discussion

Patients with R/R AML have a median overall survival of 5.3 months from the time of R/R diagnosis, and the 5-year survival rate is 12.6% (48). There is a medical need for more effective and

personalized therapy options for patients with R/R AML. Treatment resistance in AML is driven by patient-specific genetically and nongenetically modified cell subpopulations, each of which may be associated with a distinct transcriptomic state and cellular phenotype, such as drug resistance. These dynamic transcriptomic states evolve over time throughout disease progression and treatment regimens. Single-cell approaches that learn such dynamic cell state changes between the drug-sensitive (e.g., diagnosis) and resistance (R/R) states of cancer cells, combined with predictive models that identify patient-tailored treatments capable of reversing these changes, would offer the possibility of identifying effective treatments for individual patients in the R/R stage. In this study, we developed and experimentally tested a systematic approach that learns the unique transcriptomic changes between the paired diagnosis and R/R samples in patients with AML to identify

personalized, safe, and effective combinatorial therapies that selectively coinhibit the patient-specific resistant cancer cells. Selectivity is important to avoid severe coinhibition of normal cells in the relapsed stage and to deprioritize broadly toxic combinations with the aim of improving both combination efficacy and avoiding adverse drug effects. To the best of our knowledge, this is the first systematic approach to prioritize selective drug combinations for relapsed disease using the patient-specific transcriptomic and drug sensitivity history of disease progression.

Using the flow cytometry-based cell population assay, we experimentally confirmed that the predicted combinations led to improved synergistic effects in the relapsed stage, even in the absence of synergy in the diagnosis stage, hence identifying relapse-specific vulnerabilities. We further confirmed that all the predicted combinations resulted in less than 20% inhibition of normal cells, and three out of the four combinations had lower toxicity in the relapsed stage when compared with the diagnosis stage. The demonstrated accuracy at predicting synergistic and selective drug combinations supports the translational value of our model for patient-tailored combination predictions. Given that drug combination synergy is a rare phenomenon, with the estimated likelihood of synergy being around 5% in large-scale combination screens in cell lines (22, 49), and arguably even less when identifying selective synergy, our model offers a fast and accurate preclinical prioritization of safe and effective drug combinations, which can further be tested for pharmacokinetics and *in vivo* safety. We note that due to the scarce availability of bone marrow cells for flow cytometry testing, we could validate predictions only in one of the patients *ex vivo* using the primary cells from the diagnosis and relapse samples. However, our initial results in the VenEx trial samples demonstrated that the approach has predictive power for clinical outcomes, suggesting it could be used for patient selection for the venetoclax-azacitidine combination therapy (Fig. 6). Due to the complexity of clinical trial results and potential confounding factors, further validation is needed in larger patient populations and by testing the predictions in patient-derived xenograft models.

For personalized cancer treatment, we need systematic and widely applicable strategies for designing combinatorial treatments that target heterogeneous, drug-resistant cell populations. Although isolating the patient samples for a particular cell population using flow cytometry can identify drugs that selectively inhibit the target cell population (e.g., AML blast cells; refs. 12, 14), enriching for each subpopulation of both cancer and normal cells required for differential response analyses before scRNA-seq is not amenable to routine clinical application. We have demonstrated that integrated approaches based on nonenriched profiling of both single-cell transcriptomes and drug sensitivity in the whole sample composition provide a relatively unbiased and practical platform for translational discoveries in hematological malignancies, informing both monotherapy regimens and combination discoveries (20, 21). However, due to the cell population unspecificity of the bulk viability CTG drug assay, the predictions identified using the machine learning approach need to be validated with higher-precision flow cytometry population assays to confirm that only the most synergistic and selective combinations against cancer cells will be suggested as patient-tailored treatment options. The precision of our approach to selectively target AML blast cells may be further enhanced in applications in which the flow cytometry-based drug viability assessments can be routinely conducted for all AML blast subpopulations and all the drugs tested (e.g., VenEx trial). Even though the use of bone marrow aspirates is a standard practice in AML diagnosis and management (50), peripheral blood samples would be more convenient for the patients and for diagnostic purposes.

We note that some of the identified combinations may be challenging to directly translate to clinical use. For instance, the combination of BX-912 and seliciclib requires further validation because BX-912 has not yet been tested in clinical studies, similar to any other reasonably selective PDK1 inhibitors, and seliciclib has shown dose-limiting toxicities in a phase I clinical study in solid tumors (51). Future applications should focus on more selective CDK inhibitors to bypass the toxicity issues. In general, dose adjustments and further safety evaluations need to be performed in larger patient populations or using animal models. Related to preclinical toxicity evaluation, the choice of normal cells for preclinical assessment of drug-induced toxic effects in hematological cancers is not well defined, as the origin of efficacy and toxicity can be linked to the wide off-target activity of drugs (52). We used T and NK cells as non-cancerous cell types in this study because (i) these lymphocytes are considered normal cells in myeloid leukemia and (ii) they can be robustly identified using established surface markers in the complex AML samples (29). However, we note that drug-induced toxicities are not solely derived from the activity of drugs on lymphocytes or even on the cells of the hematopoietic compartment, and the measured inhibition of T and NK cells neglects, for instance, renal toxicity or cardiotoxicity. The choice of normal cell types should be made based on the disease application and drug class for the preclinical assessment of combinations in terms of both efficacy and toxicity. Before clinical translation, the predicted combinations need to be further tested in more advanced models for toxicity assessment in relevant doses and organs.

Regardless of its limitations, the predictive approach can prove highly valuable for suggesting personalized combinations for patients with R/R AML. The model predictions identified patient-tailored combinations that cotarget multiple driver proteins and signaling pathways (Supplementary Fig. S15A and S15B), which is expected to reduce therapy resistance and lead to enhanced treatment outcomes in patients with advanced R/R disease. The machine learning model enables confident identification of selective and synergistic drug combinations, along with potential molecular markers explaining the combination synergy (Supplementary Fig. S13). For instance, we identified several target genes in the AKT signaling pathway associated with the selective coinhibition effect of the sirolimus and MK-2206 combination on cancer cells (Fig. 5), which warrant further testing as predictive biomarkers in larger cohorts of patients with AML. Such expression signatures in specific cell populations may provide prognostic signatures for predicting the likelihood of therapy responses using scRNA-seq data or computational deconvolution of bulk RNA-seq that is still more accessible in large-scale clinical studies. Furthermore, predictive markers from *ex vivo* responses and single-cell RNA profiles could predict in advance a potentially safe and effective combination for patients with a higher risk of future relapse. The 2-week turnaround time makes the approach clinically actionable for AML management. Given the continuous development of sample biobanking, single-cell profiling, and data analysis capabilities in many cancer centers, our predictive approach has the potential to become clinically widely applicable to identify low-toxic and synergistic combinations for relapsed patients in a personalized manner.

Authors' Disclosures

Y. Chen reports grants from the China Scholarship Council, K. Albin Johansson's Stiftelse, and Cancer Foundation Finland during the conduct of the study. L. He reports grants from the Orion Research Foundation and the Osk. Huttunen Foundation during the conduct of the study. H. Kuusanmäki reports

personal fees from Faron Pharmaceuticals and grants from AbbVie outside the submitted work. M. Kontro reports personal fees from Faron Pharmaceuticals, Ferring Ventures, Proteina, AbbVie, Astellas, Servier, and Immedica outside the submitted work. C.A. Heckman reports grants from the Research Council of Finland, Cancer Foundation of Finland, and Sigrid Jusélius Foundation during the conduct of the study as well as grants from Novartis, Oncopeptides, Zentaris Pharmaceuticals, Kronos Bio, and WNTresearch and personal fees from Amgen and Autolus outside the submitted work. No disclosures were reported by the other authors.

Authors' Contributions

Y. Chen: Conceptualization, resources, data curation, software, formal analysis, supervision, funding acquisition, investigation, visualization, methodology, writing—original draft. **L. He:** Conceptualization, data curation, supervision. **A. Ianevski:** Conceptualization, software, supervision, validation, investigation, methodology. **K. Nader:** Software, formal analysis. **T. Ruokoranta:** Validation, investigation. **N. Linnavirta:** Validation, investigation. **J.J. Miettinen:** Resources, data curation. **M. Vähä-Koskela:** Resources, data curation. **I. Vääntinen:** Resources, data curation. **H. Kuusanmäki:** Resources, supervision, validation, methodology. **M. Kontro:** Resources, funding acquisition. **K. Porkka:** Resources, funding acquisition. **K. Wennerberg:** Supervision, validation, investigation, writing—review and editing. **C.A. Heckman:** Resources, supervision, funding acquisition, writing—review and editing. **A.K. Giri:** Data curation, supervision, investigation, writing—original draft. **T. Aittokallio:** Conceptualization, resources, supervision, funding acquisition, writing—original draft, writing—review and editing.

Acknowledgments

We are most grateful to the patients and their families for participating in the studies. We thank the Finnish Hematology Registry and Clinical Biobank for providing the AML biospecimens and associated patient data. The drug assays

were carried out at the Institute for Molecular Medicine Finland High Throughput Biomedicine Unit, hosted by the University of Helsinki and supported by HiLIFE and Biocenter Finland. scRNA-seq of the patient samples was performed at the Institute for Molecular Medicine Finland Single-Cell Analytics Unit supported by HiLIFE and Biocenter Finland. We thank CSC Finland – IT Center for Science for computational resources. We thank Minna Suvela for the preparation of the samples of patients with AML for scRNA-seq, Dr. Imre Vástrik for help with the data of patients with AML, Dr. Jie Bao for help with the flow cytometric data, and Drs. Esa Pitkänen and Harri Itkonen for many useful comments on the manuscript and figures. Supplementary Fig. S5 was created with BioRender.com. M. Kontro: The Foundation for the Finnish Cancer Institute, Cancer Foundation Finland, Finnish Medical Foundation. C.A. Heckman: Research Council of Finland (grants 334781, 320185, 352265, and 357686), Cancer Foundation Finland, Sigrid Jusélius Foundation, and Novartis. T. Aittokallio: Research Council of Finland (grants 340141, 344698, and 345803) under the frame ERA PerMed (as part of the JAK-STAT-TARGET and CLL-CLUE consortia), Novo Nordisk Foundation Interdisciplinary Synergy Programme 2021 (grant NNF21OC0070381), Norwegian Health Authority South-East (grants 2020026 and 2023105), the Cancer Society of Finland, the Norwegian Cancer Society (grants 216104 and 273810), the Sigrid Jusélius Foundation, and iCAN – Digital Precision Cancer Medicine Flagship (iCAN-MULTIDRUG). L. He: Orion Research Foundation, Cancer Foundation Finland, The Osk. Huttunen Foundation. Y. Chen: China Scholarship Council, K. Albin Johanssons Stiftelse, Cancer Foundation Finland. T. Ruokoranta: Government Research Funding/Helsinki University Hospital.

Note

Supplementary data for this article are available at Cancer Research Online (<http://cancerres.aacrjournals.org/>).

Received October 15, 2024; revised February 14, 2025; accepted May 6, 2025; posted first May 12, 2025.

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